Lecture 6: Maximum Likelihood Estimation

Math for Data Science · 2026

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Agenda

- Part I: Limit Theorems
- Part II: The Beta Distribution
- Part III: The Likelihood Function
- Part IV: MLE for the Binomial
- Part V: MLE for the Poisson
- Part VI: MLE for the Normal

Part I: Limit Theorems

The Law of Large Numbers

Assume we have i.i.d. r.v.s $X_1, X_2, X_3, \ldots$ with mean μ and variance σ^2. We take a finite sample of size n and define the sample mean as:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

This sample mean is itself a random variable. What are its expectation and variance?

$$E(\bar{X}_n) = E\left(\frac{1}{n}(X_1 + \cdots + X_n)\right) = \frac{1}{n}\big(E(X_1) + \cdots + E(X_n)\big) = \mu$$

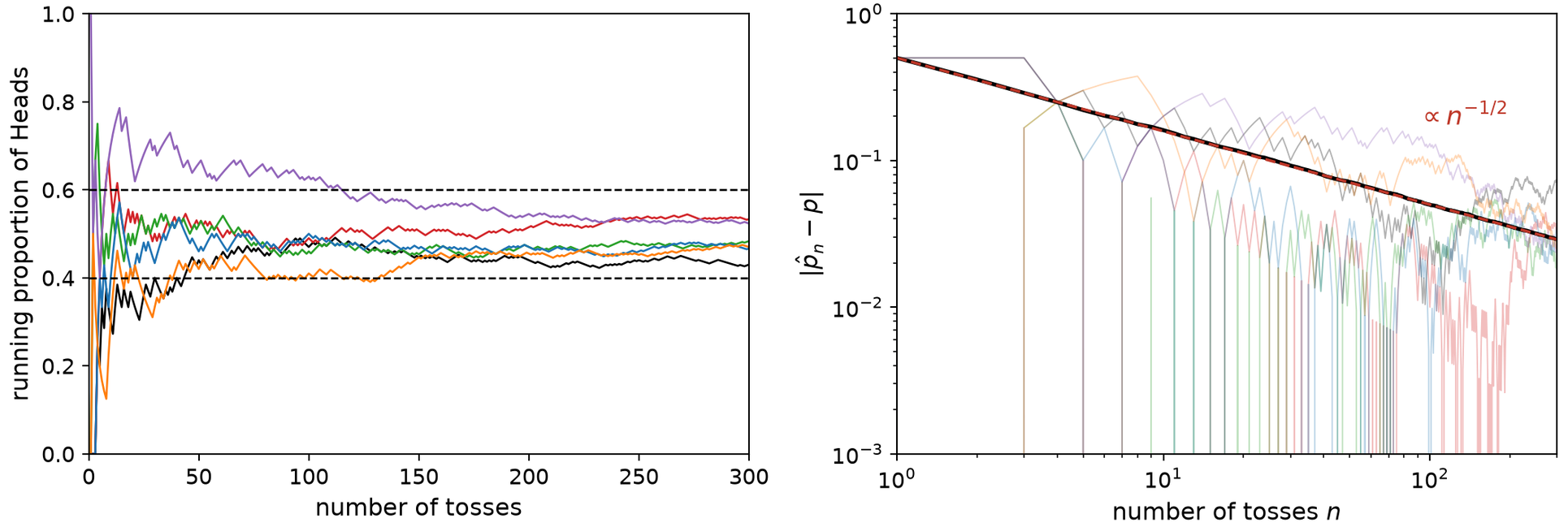
$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\,\mathrm{Var}(X_1 + \cdots + X_n) = \frac{\sigma^2}{n}$$

The Law of Large Numbers

The Law of Large Numbers states that as n grows large, the sample mean $\bar{X}_n$ converges to the true mean μ.
Convergence could mean:

- Pointwise convergence: $\bar{X}_n \to \mu$ as $n \to \infty$ (Strong Law of Large Numbers)
- Convergence in probability, written $\bar{X}_n \xrightarrow{p} \mu$: for all $\varepsilon > 0$, $P(|\bar{X}_n - \mu| > \varepsilon) \to 0$ as $n \to \infty$ (Weak Law of Large Numbers)

The Law of Large Numbers



Left: dashed lines at 0.4 and 0.6 for reference - as the number of tosses increases, the proportion of Heads approaches $1/2$. Right: the same runs as $|\hat{p}_n - 1/2|$ on log-log axes, with their root-mean-square in black - the error falls like $n^{-1/2}$, so four times the tosses buys half the error.

The Law of Large Numbers

Note: the coin is “memoryless,” meaning that after a long string of Heads, you are not more likely to get Tails; the probability of getting Tails is always $\frac{1}{2}$.

This fact does not contradict the LLN. Convergence takes place through swamping: past tosses are “swamped” by the infinitely many tosses yet to come.

The Central Limit Theorem

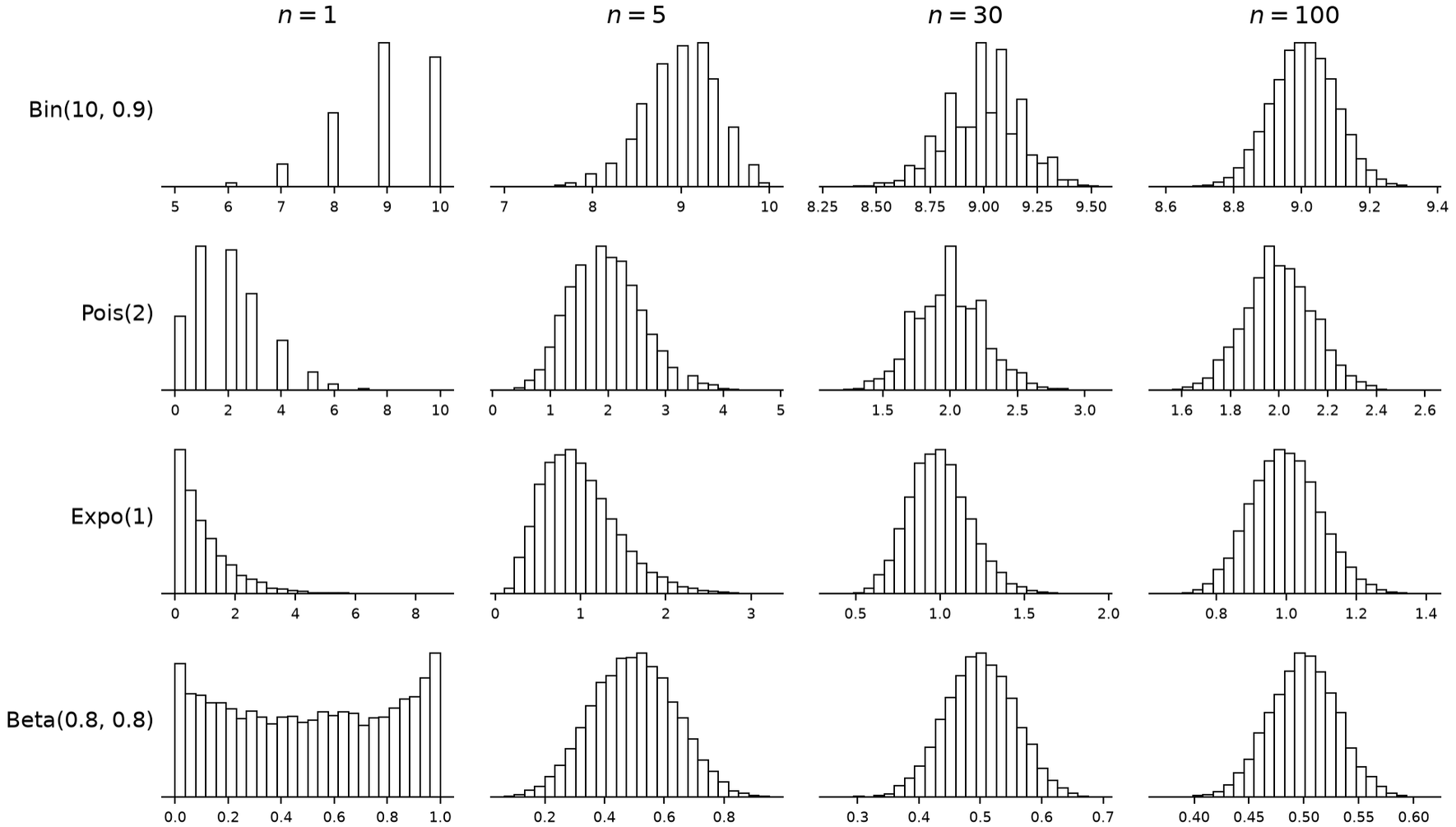
First, let's standardize our sample mean $\bar{X}_n$, recalling that this means subtracting its expectation (μ) and dividing by its standard deviation ($\sqrt{\sigma^2/n} = \sigma/\sqrt{n}$):

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right)$$

The Central Limit Theorem states that this quantity, the standardized sample mean, converges in distribution to the standard Normal as $n \to \infty$, written $\xrightarrow{d}$:

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \xrightarrow{d} \mathcal{N}(0, 1)$$

This holds regardless of the distribution of the X's as long as the mean and variance are finite (see: the evil Cauchy).



Regardless of the starting distribution of the X_i, the distribution of $\bar{X}_n$ approaches a Normal distribution as n grows.

Further Notes on the CLT

- You can un-standardize the sample mean to restate the CLT as:

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \text{ as } n \to \infty$$

- The CLT works just as well for the sum rather than the mean:

$$\sum_{i=1}^{n} X_i \xrightarrow{d} \mathcal{N}\left(n\mu, n\sigma^2\right) \text{ as } n \to \infty$$

- While the CLT works for any distribution with finite mean and variance, the underlying distribution matters for how large n has to be before the Normal approximation starts to look accurate
- As a shortcut, you can write the CLT in approximation form: $\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$

Normal Approximation to the Binomial

Recalling that the Binomial(n, p) is the sum of n Bernoullis with probability p, we can even use the Normal distribution to approximate the Binomial. Since from the previous slide,

$$\sum_{i=1}^{n} X_i \approx \mathcal{N}\left(n\mu, n\sigma^2\right)$$

and recalling that the mean of a Bernoulli is p and its variance is $p(1-p)$, we can use the CLT to say:

$$\sum_{i=1}^{n} X_i \approx \mathcal{N}\big(np,\, np(1-p)\big)$$

Part II: The Beta Distribution

The mysterious black box

- Sometimes you get two quarters
- Sometimes the quarter just disappears
- What is the probability of getting two quarters?

- You lose on the first trial.
- You try once more and win.
- You guess naively that $P(🟡🟡) = \frac{1}{2}$

You use the remaining quarters, ending at 14 wins and 27 losses in 41 trials

- Now you guess that $P(🟡🟡) = \frac{14}{41}$



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Will Kurt, *Bayesian Statistics the Fun Way*, Ch. 5

The mysterious black box

We have two hypotheses for the probabilities:

- H_1: $P(🟡🟡) = \frac{1}{2}$
- H_2: $P(🟡🟡) = \frac{14}{41}$

How probable is what we observed if H_1 is true, versus if H_2 is true?



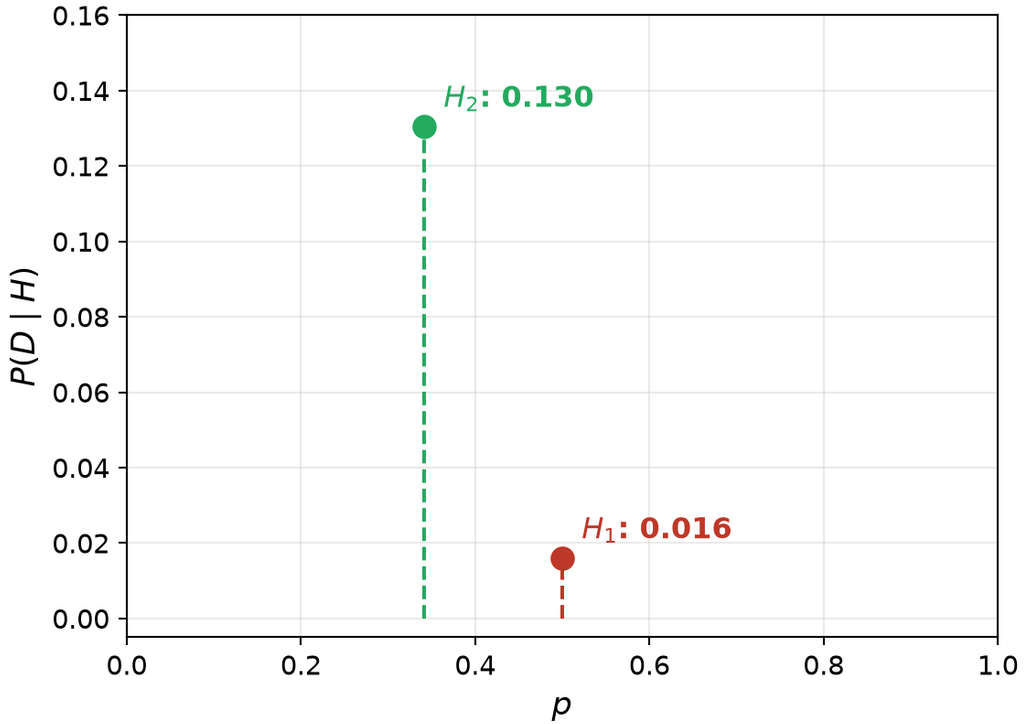
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Calculating the probabilities

Assume each try is independent and wins with the same probability $p = P(🟡🟡)$, the value the hypothesis proposes. The number of wins, $k = 14$ in $n = 41$ tries, then follows the binomial distribution from before:

$$P(D \mid H) = \mathcal{B}(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k} = \binom{41}{14} p^{14} (1-p)^{27}$$

hypothesis	$p(🟡🟡)$	$P(D \mid H)$
H_1	$\frac{1}{2}$	≈ 0.016
H_2	$\frac{14}{41}$	≈ 0.130

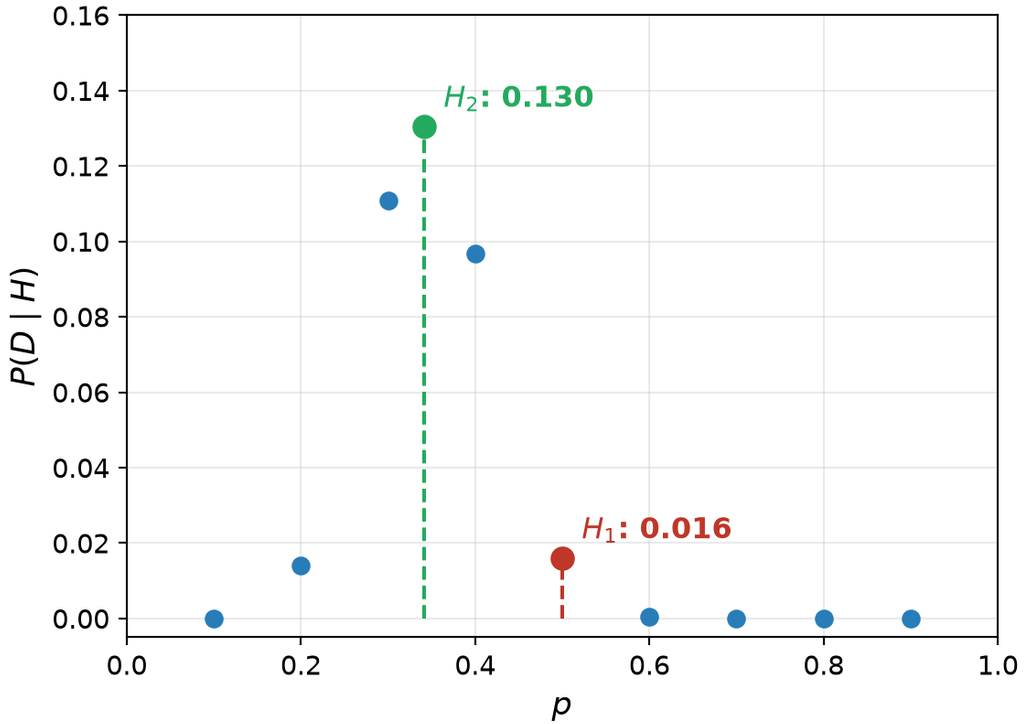


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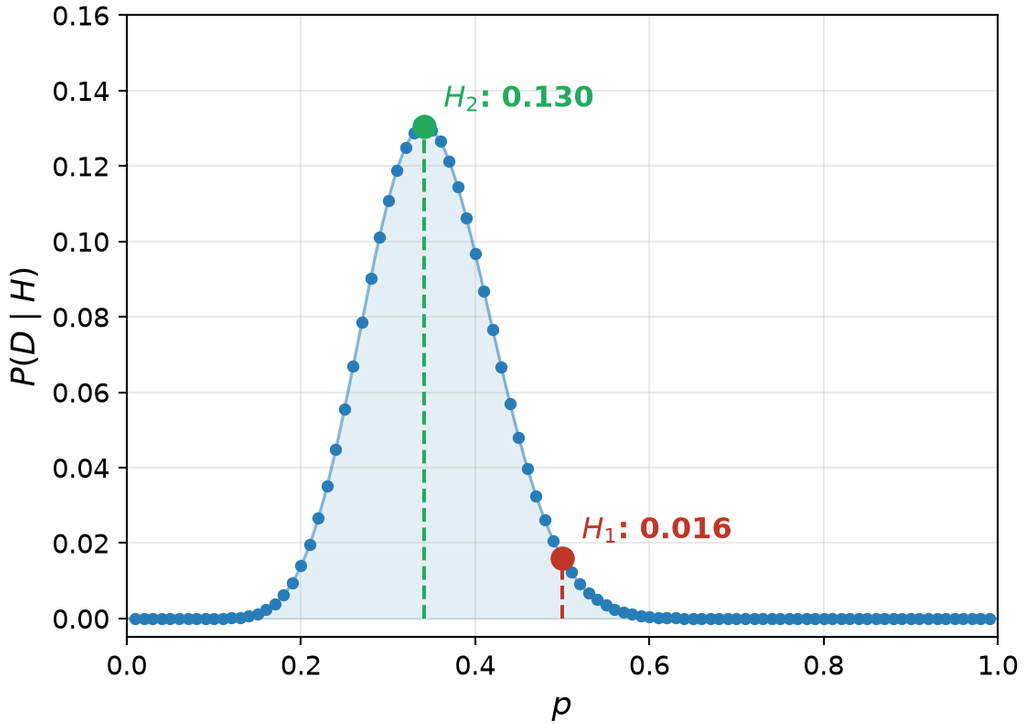
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Evaluating $\mathcal{B}(14; 41, p)$ at more and more values of p traces out a continuous curve. This is the likelihood function $\mathcal{L}(p)$ for our data.



The Beta distribution

That continuous curve has the shape of a Beta distribution:

$$f(p \mid \alpha, \beta) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{\mathcal{N}(\alpha, \beta)}$$

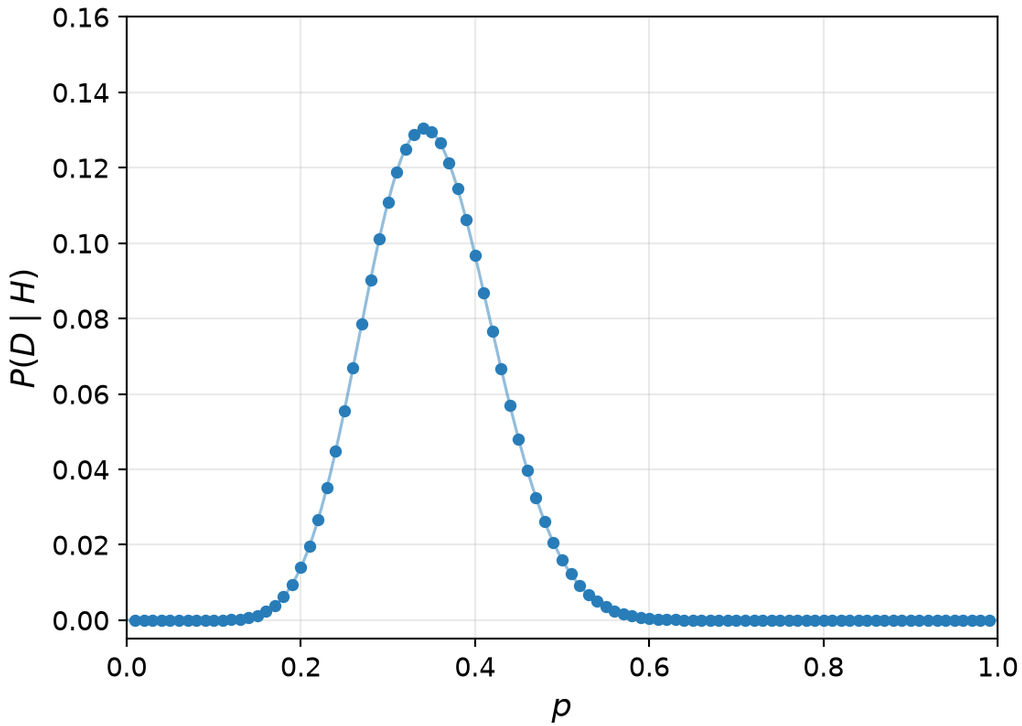
with normalizing constant

$$\mathcal{N}(\alpha, \beta) = \int_0^1 p^{\alpha-1}(1-p)^{\beta-1}\, dp.$$

Matching exponents with $p^k(1-p)^{n-k}$:

$$\begin{aligned} \alpha - 1 &= k, & \beta - 1 &= n - k \\ \Rightarrow \quad \alpha &= k + 1 = 15, & \beta &= n - k + 1 = 28 \end{aligned}$$

Our likelihood is proportional to Beta$(15, 28)$: same shape, different scale.



Part III: The Likelihood Function

Maximum Likelihood Estimation

The most common method for estimating parameters in a parametric model is the maximum likelihood method.

- Our first example of statistical learning!
- Parametric, in the sense that we specify a distribution and wish to estimate its parameters
- Frequentist, in the sense that these parameters are fixed and the data is random

The Likelihood Function

Let $X_1, \ldots, X_n$ be i.i.d. with PDF or PMF $f(x;\theta)$, and let $x_1, \ldots, x_n$ be the values we observed. We write θ for the unknown parameter; for the black box, θ is p.

Definition. The likelihood function is

$$\mathcal{L}(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

With the data fixed we write $\mathcal{L}_n(\theta)$ for short. The log-likelihood function is $\ell_n(\theta) = \log \mathcal{L}_n(\theta)$.

The likelihood function is just the joint density of the data - except that we treat it as a function of the parameter θ, with the data held fixed.

The Maximum Likelihood Estimator

Definition. The maximum likelihood estimator (MLE), denoted $\hat{\theta}_n$, is the value of θ that maximizes $\mathcal{L}_n(\theta)$.

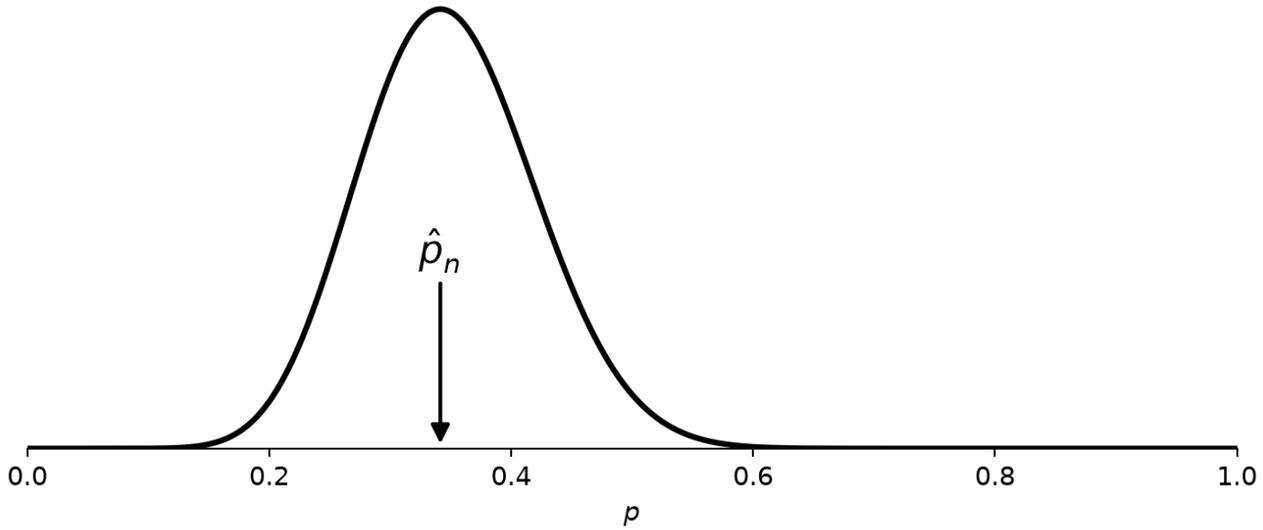
To find it: differentiate, set the derivative to zero, solve, and check the second derivative.

$$\left.\frac{d\,\ell_n(\theta)}{d\theta}\right|_{\theta=\hat{\theta}_n} = 0$$

What the Likelihood Looks Like

Apply the definition to the black box - 41 Bernoulli trials, observed outcomes $x_i \in \{0, 1\}$ with $\sum_i x_i = 14$ successes:

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{41} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{14}(1-\theta)^{27} \;\propto\; \mathrm{Beta}(15,\, 28)$$

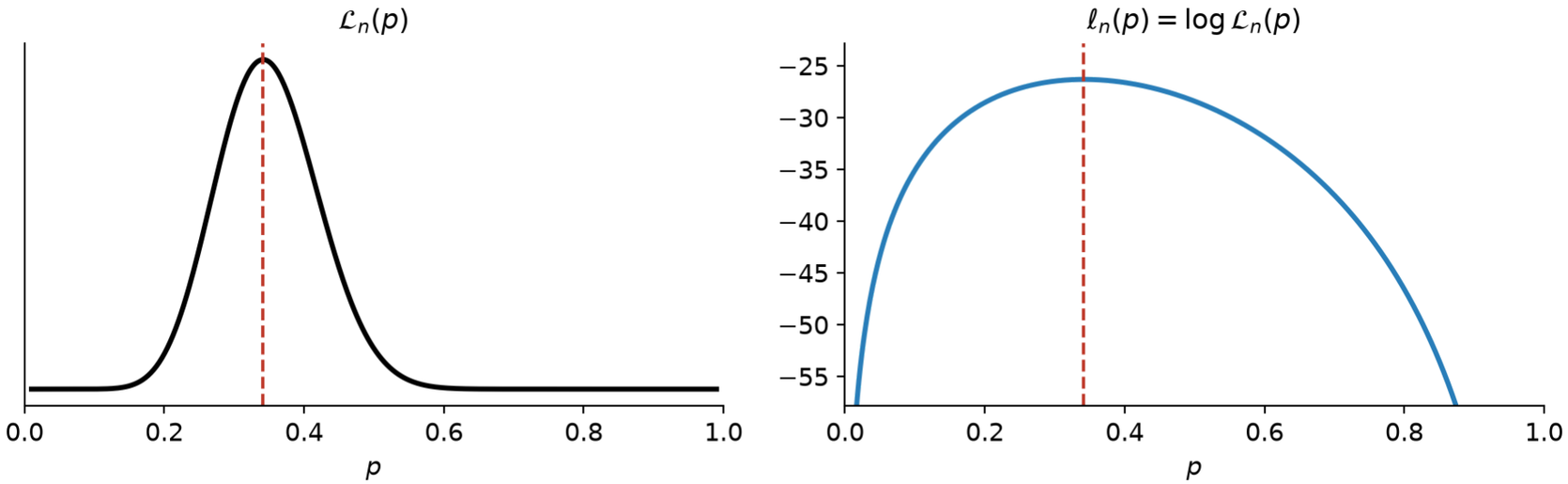


Its peak is the MLE, $\hat{\theta}_n = 14/41 \approx 0.34$.

Unlike a PDF, a likelihood does not have to integrate to one. Multiplying it by a constant changes the scale, not the position of the maximum.

Why We Maximize the Log-Likelihood

The maximum of $\ell_n(\theta)$ occurs at the same place as the maximum of $\mathcal{L}_n(\theta)$ - because $\log$ is strictly increasing on the positive reals.



Different shape, same argmax - and sums are far easier to differentiate than products.

Part IV: MLE for the Binomial

How Good Is a Goalkeeper?

A goalkeeper faces 5 shots on goal in every game, and saves each one independently with the same unknown probability θ. Then X_i, the number of saves in game i, is i.i.d. with

$$X_i \sim \text{Binomial}(5, \theta)$$

where θ stands in for the Binomial's p. Its likelihood is the probability of the saves we observed, as a function of θ:

$$\mathcal{L}_n(\theta) = P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n; \theta)$$

Four Games: the Data

Start with a small dataset of only four games, 5 shots in each:

- In the first game, the goalie had 1 save
- In the second, 3 saves
- In the third and fourth, 2 saves each

Our next task: write down the likelihood of this dataset, recalling that the PMF of the Binomial with 5 trials is

$$P(X_i = x) = \binom{5}{x} \theta^x (1-\theta)^{5-x}$$

Four Games: Write the Likelihood

The X_i are identically distributed, so each follows the same $\text{Binomial}(5, \theta)$ with the same θ. We can write the probability of each X_i as

$$P(X_1 = 1) = \binom{5}{1}\theta^1(1-\theta)^{5-1}$$

$$P(X_2 = 3) = \binom{5}{3}\theta^3(1-\theta)^{5-3}$$

$$P(X_3 = 2) = P(X_4 = 2) = \binom{5}{2}\theta^2(1-\theta)^{5-2}$$

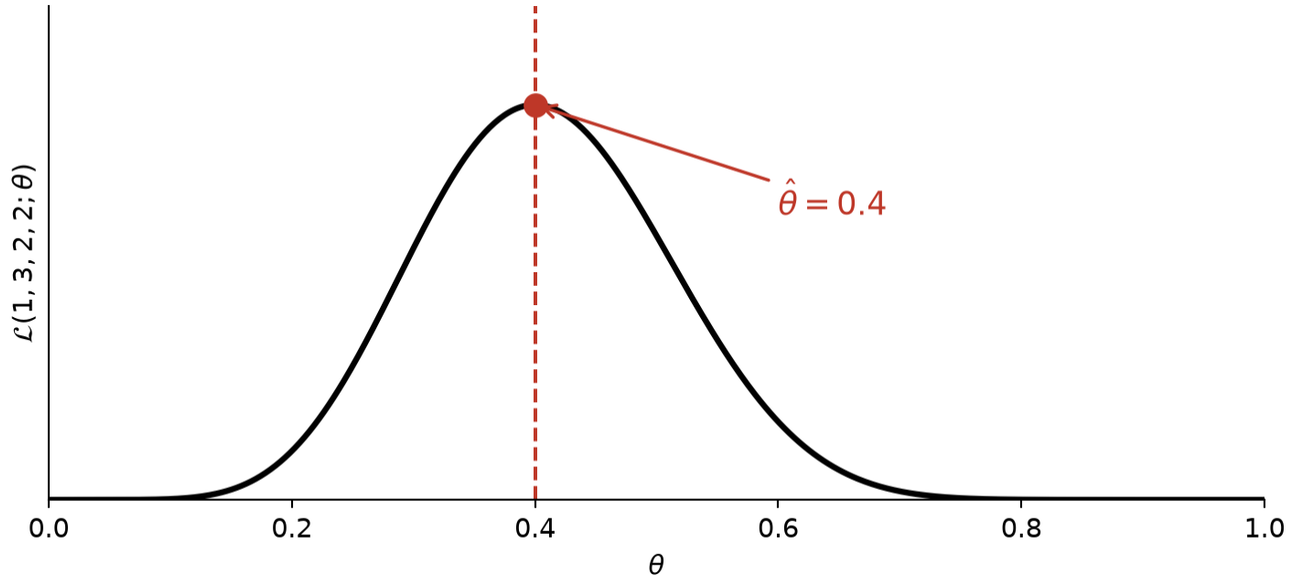
Four Games: Write the Likelihood

The X_i are also independent, so the joint probability of observing them all is the product of the individual probabilities:

$$
\begin{aligned}
P(X_1 = 1, X_2 = 3, X_3 = 2, X_4 = 2) &= \binom{5}{1}\theta^1(1-\theta)^4 \times \binom{5}{3}\theta^3(1-\theta)^2 \times \left(\binom{5}{2}\theta^2(1-\theta)^3\right)^2 \\
&= \binom{5}{1}\binom{5}{3}\binom{5}{2}\binom{5}{2}\theta^8(1-\theta)^{12} \\
&= (5 \cdot 10 \cdot 10 \cdot 10)\,\theta^8(1-\theta)^{12} \\
&\propto \theta^8(1-\theta)^{12}
\end{aligned}
$$

The binomial coefficients collapse into a constant - it does not depend on θ, so it cannot move the maximum.

Four Games: the Function We Are Maximizing



$\mathcal{L}_n(\theta) \propto \theta^8(1-\theta)^{12}$. We are looking for the peak - let's find it with calculus rather than by eye.

Four Games: Differentiate and Solve

Write $\mathcal{L}_n(\theta) = c\,\theta^8(1-\theta)^{12}$, where the constant c collects the binomial coefficients. By the constant multiple rule, the product rule, and the chain rule:

$$\begin{aligned}
\frac{d}{d\theta}\mathcal{L}_n(\theta) &= c\Big(8\theta^7(1-\theta)^{12} + \theta^8\big(12(1-\theta)^{11}\big)(-1)\Big) \\
&= c\Big(8\theta^7(1-\theta)^{12} - 12\theta^8(1-\theta)^{11}\Big) \\
&= 4c\,\theta^7(1-\theta)^{11}\Big(2(1-\theta) - 3\theta\Big) \\
&= 4c\,\theta^7(1-\theta)^{11}\Big(2 - 5\theta\Big)
\end{aligned}$$

Setting this to zero gives $\theta = 0$, $\theta = 1$ or $\theta = 2/5$. The likelihood is zero at 0 and at 1, so its maximum is at $\hat{\theta} = 2/5$ (check: $\mathcal{L}_n''(2/5) < 0$).

Four Games: Interpret the Result

$$\hat{\theta} = \frac{8}{20} = 0.4$$

The MLE for θ is just the total number of successes $(1 + 3 + 2 + 2 = 8)$ over the total number of trials $(4 \times 5 = 20)$.

Exactly what you would have guessed without any calculus. That is reassuring.

Step 1: Write the Likelihood

Now in general: n games, where game i has y_i attempts and x_i saves. The games are still independent, but with different numbers of attempts they are no longer identically distributed: $X_i \sim \text{Binomial}(y_i, \theta)$. The likelihood is still a product:

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} \binom{y_i}{x_i} \theta^{x_i} (1-\theta)^{y_i - x_i}$$

Differentiating a product of n terms with the product rule is not something anyone wants to do. So we take the log first.

Log Rules

Taking the log turns the product into a sum. It may be helpful to recall the following facts:

- $\log(ab) = \log(a) + \log(b)$
- $\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$
- $\log(\prod_{i=1}^{n} a_i) = \sum_{i=1}^{n} \log(a_i)$
- $\log a^b = b \log a$

Step 2: Take the Log

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} \binom{y_i}{x_i} \theta^{x_i} (1-\theta)^{y_i - x_i}$$

$$\ell_n(\theta) = \sum_{i=1}^{n} \left(\log \binom{y_i}{x_i} + x_i \log \theta + (y_i - x_i) \log(1-\theta) \right)$$

Note that $\log \binom{y_i}{x_i}$ carries no θ, so its derivative is zero.

Step 3: Differentiate

$$\begin{aligned}\frac{d\ell_n}{d\theta} &= \sum_{i=1}^{n} x_i \left(\frac{1}{\theta}\right) + \sum_{i=1}^{n} (y_i - x_i) \left(\frac{1}{1-\theta}\right)(-1) \\ &= \sum_{i=1}^{n} x_i \left(\frac{1}{\theta}\right) - \sum_{i=1}^{n} (y_i - x_i) \left(\frac{1}{1-\theta}\right)\end{aligned}$$

Step 4: Solve

Set the derivative to zero and solve for $\hat{\theta}$:

$$
\begin{aligned}
0 &= \sum_{i=1}^{n} x_i \left(\frac{1}{\hat{\theta}}\right) - \sum_{i=1}^{n} (y_i - x_i) \left(\frac{1}{1-\hat{\theta}}\right) \\
\sum_{i=1}^{n} (y_i - x_i) \left(\frac{1}{1-\hat{\theta}}\right) &= \sum_{i=1}^{n} x_i \left(\frac{1}{\hat{\theta}}\right) \\
\hat{\theta} \sum_{i=1}^{n} (y_i - x_i) &= (1-\hat{\theta}) \sum_{i=1}^{n} x_i \\
\hat{\theta} \sum_{i=1}^{n} y_i - \hat{\theta} \sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} x_i - \hat{\theta} \sum_{i=1}^{n} x_i \\
\hat{\theta} \sum_{i=1}^{n} y_i &= \sum_{i=1}^{n} x_i \quad \longrightarrow \quad \hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}
\end{aligned}
$$

Interpret the Result

$$\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}$$

Again the MLE for θ is the total number of successes over the total number of trials. This should sound somewhat intuitive!

And it matches our four-game answer: there every game had $m = 5$ attempts, so $\sum y_i = mn = 20$ and $\sum x_i = 8$, giving $\hat{\theta} = 8/20$. ✓

Part V: MLE for the Poisson

How Many Shots per Game?

In the four-game example every game had 5 shots. In general the number varies: over five games our goalkeeper faced 3, 5, 4, 2 and 6 shots.

A count of events in a fixed interval - here, shots in one game - is often modelled with a Poisson distribution. Its PMF with parameter λ is

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots$$

Same recipe as before, four steps: write the likelihood, log it, differentiate it, solve.

Step 1: Write the Likelihood

Given a random sample $X_1, X_2, \ldots, X_n$ from a Poisson with parameter λ, with observed values $x_1, \ldots, x_n$, the likelihood function is

$$\mathcal{L}_n(\lambda) = \prod_{i=1}^{n} P(X_i = x_i) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

Step 2: Take the Log

Take the logarithm of the full likelihood:

$$\ell_n(\lambda) = \log \mathcal{L}_n(\lambda) = \log\left(\prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}\right)$$

Using $\log(AB) = \log A + \log B$ and $\log(A/B) = \log A - \log B$:

$$\ell_n(\lambda) = \sum_{i=1}^{n} \log\left(\frac{\lambda^{x_i} e^{-\lambda}}{x_i!}\right) = \sum_{i=1}^{n} \left(x_i \log \lambda - \lambda - \log(x_i!)\right)$$

Step 3: Differentiate

$$\frac{d\ell_n(\lambda)}{d\lambda} = \sum_{i=1}^{n} \left(\frac{x_i}{\lambda} - 1 \right)$$

The $\log(x_i!)$ term is a constant in λ, so it vanishes - exactly as $\log\binom{y_i}{x_i}$ did in the Binomial case.

Step 4: Solve

Set the derivative equal to zero to find the critical point:

$$\sum_{i=1}^{n} \left(\frac{x_i}{\hat{\lambda}} - 1 \right) = 0$$

Multiply through by $\hat{\lambda}$:

$$\sum_{i=1}^{n} x_i - n\hat{\lambda} = 0$$

And solve:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Interpret the Result

The MLE for the parameter λ of the Poisson distribution is the sample mean:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Which makes sense: λ is the mean of a Poisson. The MLE recovers the obvious estimator - as it did for the Binomial.

For our five games: $\hat{\lambda} = (3 + 5 + 4 + 2 + 6)/5 = 4$ shots per game.

Check: the Second Derivative

To be sure the critical point is a maximum and not a minimum, check the second derivative:

$$\frac{d^2\ell_n(\lambda)}{d\lambda^2} = -\sum_{i=1}^{n} \frac{x_i}{\lambda^2}$$

This is negative for all $\lambda > 0$, so the log-likelihood is concave - confirming that the solution we found is indeed a maximum.

Part VI: MLE for the Normal

Two Parameters at Once

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$, with both parameters unknown. The PDF is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right)$$

Now $\theta = (\mu, \sigma^2)$ is a vector. The recipe is unchanged; we just set two partial derivatives to zero.

Steps 1-2: Write the Likelihood, Take the Log

$$\mathcal{L}_n(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Taking logs turns the product into a sum and cancels the exponential outright:

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

This is the payoff for logging: $e^{(\cdot)}$ inside a product would be miserable; $\log$ of it is a quadratic.

Steps 3-4: Differentiate and Solve for $\hat{\mu}$

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Differentiate with respect to μ, holding σ^2 fixed (chain rule on the squared term):

$$\frac{\partial \ell_n}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

Set it to zero. Since $\sigma^2 > 0$, we can multiply it away:

$$\sum_{i=1}^{n}(x_i - \hat{\mu}) = 0 \implies \sum_{i=1}^{n} x_i = n\hat{\mu} \implies \boxed{\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i}$$

The MLE of the mean is the sample mean - and notably it does not depend on σ^2.

Steps 3-4: Differentiate and Solve for $\hat{\sigma}^2$

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

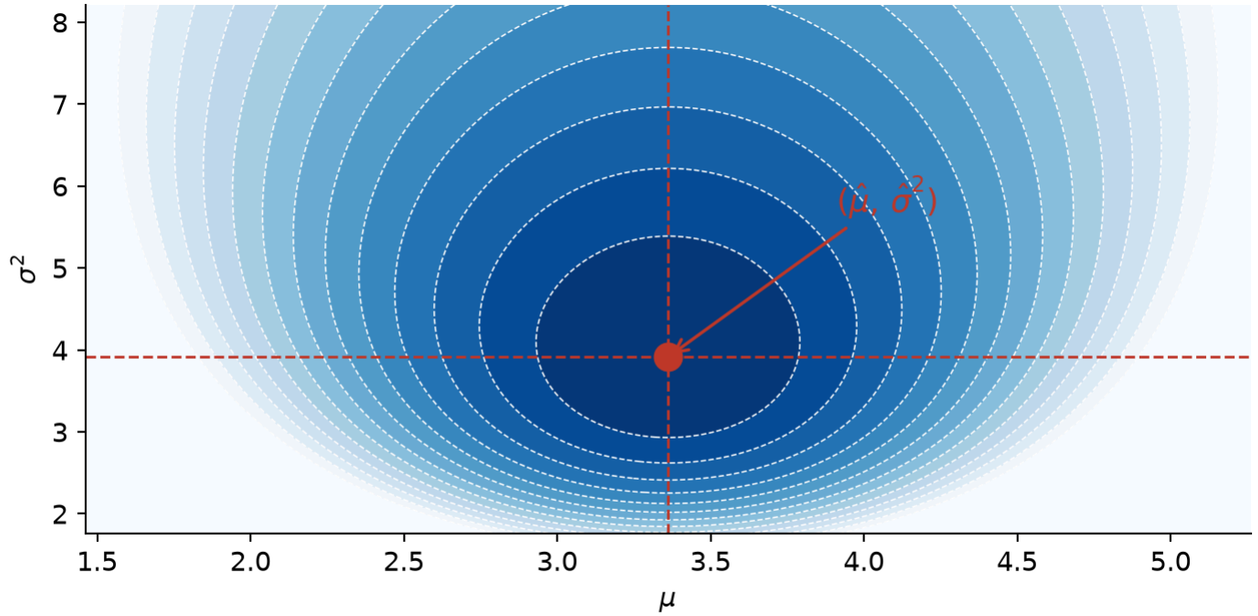
Now differentiate with respect to σ^2, treating it as a single variable $v = \sigma^2$:

$$\frac{\partial \ell_n}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Set to zero, multiply through by $2v^2$, and substitute $\hat{\mu} = \bar{x}$:

$$-n\hat{v} + \sum_{i=1}^{n}(x_i - \bar{x})^2 = 0 \implies \boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Seeing It: the Log-Likelihood Surface



One sample of $n = 40$ from $\mathcal{N}(3, 4)$. Setting both partial derivatives to zero finds the top of this hill.

Summary - maximum likelihood estimation

Concept	Key formula
LLN	$\bar{X}_n \to \mu$ as $n \to \infty$
CLT	$\sqrt{n}\,(\bar{X}_n - \mu)/\sigma \to \mathcal{N}(0,1)$
Likelihood	$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$
Log-likelihood	$\ell_n(\theta) = \sum_{i=1}^{n} \log f(x_i;\theta)$
MLE	$\hat{\theta}_n = \arg\max_\theta \mathcal{L}_n(\theta) = \arg\max_\theta \ell_n(\theta)$
Recipe	write $\mathcal{L}$ → take $\log$ → set $\ell' = 0$ → check $\ell'' < 0$

Further Reading

Kurt, *Bayesian Statistics the Fun Way* - Ch. 5 - The beta distribution

Wasserman, *All of Statistics*

- Ch. 9.1 - parametric inference and the parameter of interest
- Ch. 9.3 - the likelihood function and the MLE
- Ch. 9.4 - properties of the MLE: consistency, equivariance, asymptotic normality

Blitzstein & Hwang, *Introduction to Probability*

- Ch. 10.2–10.3 (p. 467–474) - LLN and CLT

Herman & Strang, *Calculus, Volume 2*

- Ch. 6.1, Ch. 6.3 - power series, Taylor and Maclaurin series

Optional

- Orloff & Bloom, MIT notes on MLE - math.mit.edu/~dav/05.dir/class10-prep.pdf